2026Fall Quiz

for (MATH301) Modern Algebra I

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Problem 1.1. Let G be a finite group. Suppose that G has no element of order 2. Prove that $|G|$ is odd.

Solution. For every $g \in G \setminus \{1_G\}$, we have $g \neq g^{-1}$; otherwise $g^2 = 1_G$, so g would have order 2. Hence the elements of $G \setminus \{1_G\}$ occur in disjoint pairs

$$\{g, g^{-1}\}.$$

Thus $|G| - 1$ is even, and therefore $|G|$ is odd.

□

Problem 1.2. Let $\sigma \in S_n$ have order 6. Prove that if the cycle decomposition of σ contains no 6-cycle, then it must contain both a 2-cycle and a 3-cycle.

Solution. Write the disjoint cycle decomposition of σ as

$$\sigma = c_1 c_2 \cdots c_r,$$

where the nontrivial cycles c_i have lengths m_i. Then

$$|\sigma| = \operatorname{lcm}(m_1, \ldots, m_r) = 6.$$

Hence each m_i divides 6. Since there is no 6-cycle, every m_i is 2 or 3. If there were no 2-cycle, then $|\sigma|$ would divide 3; if there were no 3-cycle, then $|\sigma|$ would divide 2. Both contradict $|\sigma| = 6$. Therefore the cycle decomposition contains both a 2-cycle and a 3-cycle.

□

Problem 2.1. Let G be a group and let $\mathcal{S}$ be the set of all subgroups of G. Show that

$$g \cdot H = gHg^{-1},$$

for $g \in G$ and $H \in \mathcal{S}$, defines a left group action of G on $\mathcal{S}$.

Solution. First, the proposed map is well defined. For $g \in G$ and $H \leq G$, we have

$$1_G = g1_G g^{-1} \in gHg^{-1}.$$

If $gh_1g^{-1}, gh_2g^{-1} \in gHg^{-1}$, then

$$(gh_1g^{-1})(gh_2g^{-1})^{-1} = g(h_1h_2^{-1})g^{-1} \in gHg^{-1},$$

since $h_1h_2^{-1} \in H$. By the Subgroup Criterion, $gHg^{-1} \leq G$.

Now let $g_1, g_2 \in G$ and $H \in \mathcal{S}$. Then

$$\begin{aligned} g_1 \cdot (g_2 \cdot H) &= g_1(g_2Hg_2^{-1})g_1^{-1} \\ &= (g_1g_2)H(g_1g_2)^{-1} \\ &= (g_1g_2) \cdot H, \end{aligned}$$

and

$$1_G \cdot H = 1_G H 1_G^{-1} = H.$$

Thus $g \cdot H = gHg^{-1}$ defines a left group action of G on $\mathcal{S}$.

□

Problem 2.2. Let $C_{12} = \langle x \rangle$ and $C_{18} = \langle y \rangle$ be cyclic groups of orders 12 and 18, respectively. Find, with proof, the number of homomorphisms

$$\varphi \colon C_{12} \to C_{18}.$$

Notation. Following MATH000, C_n denotes an abstract cyclic group of order n. The original quiz uses Z_n for the same object.

Solution. A homomorphism $\varphi \colon C_{12} \to C_{18}$ is determined by the image of the generator x. Write

$$\varphi(x) = y^a, \qquad 0 \leq a < 18.$$

Since $x^{12} = 1$, we must have

$$1 = \varphi(x^{12}) = y^{12a},$$

so

$$18 \mid 12a \quad \Longleftrightarrow \quad 3 \mid a.$$

Thus the only possible values are

$$a = 0, 3, 6, 9, 12, 15.$$

Conversely, for each such a, define

$$\varphi_a(x^k) = y^{ak}.$$

If $x^k = x^\ell$, then $12 \mid (k - \ell)$; since $18 \mid 12a$, it follows that $18 \mid a(k - \ell)$, so $y^{ak} = y^{a\ell}$. Hence φ_a is well defined, and

$$\varphi_a(x^{k+\ell}) = y^{a(k+\ell)} = y^{ak}y^{a\ell} = \varphi_a(x^k)\varphi_a(x^\ell),$$

so φ_a is a homomorphism. These six homomorphisms are distinct because they send x to distinct elements of C_{18}. Therefore the number of homomorphisms is

$$\boxed{6}.$$

□

Problem 3.1. Let G act on a nonempty set A. For $a \in A$, define the orbit of a by

$$G \cdot a = \{g \cdot a \mid g \in G\}.$$

Prove that if $b \in G \cdot a$, then $G \cdot b = G \cdot a$. Deduce that the distinct orbits form a partition of A.

Solution. Suppose that $b \in G \cdot a$, so $b = g \cdot a$ for some $g \in G$. Then

$$\begin{aligned} G \cdot b &= \{h \cdot (g \cdot a) \mid h \in G\} \\ &= \{(hg) \cdot a \mid h \in G\} \\ &= G \cdot a, \end{aligned}$$

since $\{hg \mid h \in G\} = G$.

Every $a \in A$ lies in $G \cdot a$, so the orbits cover A. If two orbits $G \cdot a$ and $G \cdot b$ intersect, choose $c \in (G \cdot a) \cap (G \cdot b)$. By the first part,

$$G \cdot a = G \cdot c = G \cdot b.$$

Hence any two distinct orbits are disjoint, and the distinct orbits form a partition of A.

□

Problem 3.2. Let G be a group and let $\varphi \in \mathrm{Aut}(G)$. Define

$$\mathrm{Fix}(\varphi) = \{g \in G \mid \varphi(g) = g\}.$$

(a) Prove that $\mathrm{Fix}(\varphi)$ is a subgroup of G.

(b) Let $G = \langle x \rangle$ be cyclic of order 12, and let φ be the automorphism determined by $\varphi(x) = x^5$. Determine $\mathrm{Fix}(\varphi)$.

Solution.

(a) Since $\varphi(1_G) = 1_G$, the set $\mathrm{Fix}(\varphi)$ is nonempty. If $a, b \in \mathrm{Fix}(\varphi)$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = ab^{-1}.$$

Thus $ab^{-1} \in \mathrm{Fix}(\varphi)$, so $\mathrm{Fix}(\varphi) \leq G$ by the Subgroup Criterion.

(b) Every element of G has the form x^k, and

$$\begin{aligned} x^k \in \mathrm{Fix}(\varphi) &\iff x^{5k} = x^k \\ &\iff x^{4k} = 1_G \\ &\iff 12 \mid 4k \\ &\iff 3 \mid k. \end{aligned}$$

Therefore

$$\mathrm{Fix}(\varphi) = \langle x^3 \rangle = \{1_G, x^3, x^6, x^9\}.$$

□

Problem 4.1. Let G be a group and let $H \leq G$. Prove that H is abelian if and only if

$$\langle H, C_G(H) \rangle = C_G(H).$$

Solution. By definition, H is abelian if and only if every element of H commutes with every element of H, which is equivalent to $H \leq C_G(H)$. Since $C_G(H)$ is a subgroup of G, we also have

$$H \leq C_G(H) \quad \iff \quad \langle H, C_G(H) \rangle = C_G(H).$$

Combining these equivalences proves the claim.

□

Problem 4.2. Let $\varphi\colon G \to H$ be a group homomorphism. Determine whether each of the following statements is true or false. For each false statement, give a counterexample.

(a) $\varphi(1_G) = 1_H$.

(b) If $A \trianglelefteq G$, then $\varphi(A) \trianglelefteq H$.

(c) If $E \leq H$, then $\varphi^{-1}(E) \leq G$.

(d) If $E \trianglelefteq H$, then $\varphi^{-1}(E) \trianglelefteq G$.

(e) If $x \in G$ has finite order, then $|x|$ divides $|\varphi(x)|$.

Solution.

(a) **True.** Since $\varphi(1_G)^2 = \varphi(1_G)$, cancellation in H gives $\varphi(1_G) = 1_H$.

(b) **False.** Define $\varphi\colon \mathbb{Z}/2\mathbb{Z} \to S_3$ by $\varphi([k]_2) = (12)^k$ and take $A = \mathbb{Z}/2\mathbb{Z}$. Although $A \trianglelefteq \mathbb{Z}/2\mathbb{Z}$, its image $\langle (12) \rangle$ is not normal in S_3, since

$$(13)\langle (12) \rangle (13)^{-1} = \langle (23) \rangle \neq \langle (12) \rangle.$$

The image of a normal subgroup is normal in $\varphi(G)$, but need not be normal in H.

(c) **True.** The set $\varphi^{-1}(E)$ contains 1_G. If $x, y \in \varphi^{-1}(E)$, then

$$\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} \in E,$$

so $xy^{-1} \in \varphi^{-1}(E)$. Thus $\varphi^{-1}(E) \leq G$.

(d) **True.** By (c), $\varphi^{-1}(E) \leq G$. For $g \in G$ and $x \in \varphi^{-1}(E)$,

$$\varphi(gxg^{-1}) = \varphi(g)\varphi(x)\varphi(g)^{-1} \in E$$

because $E \trianglelefteq H$. Hence $gxg^{-1} \in \varphi^{-1}(E)$, proving normality.

(e) **False.** The reduction homomorphism

$$\varphi\colon \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}, \qquad [k]_4 \mapsto [k]_2$$

sends the element $[1]_4$ of order 4 to $[1]_2$ of order 2, and $4 \nmid 2$. In general, the correct divisibility is $|\varphi(x)| \mid |x|$ for an element x of finite order.

□